

$$1) a) A = 720 \text{ mm}^2$$

$$l_E = 460 \text{ mm}$$

$$N = 800$$

Magnetisierungskennlinie

$$\Phi = 8,3 \cdot 10^{-4} \text{ Wb}$$

$$\text{Wb} = \text{Vs} = \text{Tm}^2$$

$$I = ?$$

$$\Phi = B \cdot A \Rightarrow B = \frac{\Phi}{A} = \frac{8,3 \cdot 10^{-4} \text{ Wb}}{720 \text{ mm}^2} = \frac{8,3 \cdot 10^{-4}}{720 \cdot (10^{-3})^2} \cdot \frac{\text{Tm}^2}{\text{m}^2} =$$

$$B = \underline{\underline{1,153 \text{ T}}}$$

$$B = \mu \cdot H$$

Da μ bei ferromagnetischen Stoffen nicht konstant ist, muss die Magnetisierungskennlinie verwendet werden.

$$H = \underline{\underline{550 \frac{\text{A}}{\text{m}}}} \text{ (aus Kennlinie)}$$

Durchflutungsgesetz: vereinfacht für „Serienhaltung“

$$\Theta = N \cdot I = \oint \vec{H} \cdot d\vec{s} = \sum H \cdot l$$

$$\Rightarrow I = \frac{H \cdot l_E}{N} = \frac{550 \frac{\text{A}}{\text{m}} \cdot 0,46 \text{ m}}{800} = \underline{\underline{0,31625 \text{ A}}}$$

b) I' für selbes Φ mit Luftspalt $l_L = 0,5 \text{ mm}$?

$$B = 1,153 \text{ T} \xrightarrow{\text{Eisen}} H_E = 550 \frac{\text{A}}{\text{m}}$$

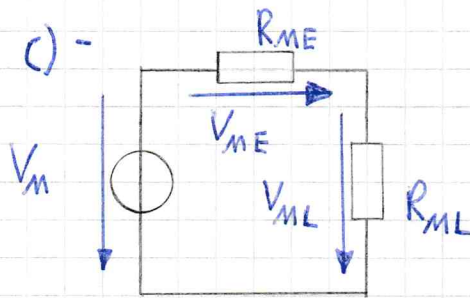
$$\Rightarrow H_L = \frac{B}{\mu_0 \cdot \mu_r} = \frac{1,153 \frac{\text{Vs}}{\text{m}^2}}{4\pi \cdot 10^{-7} \cdot 1 \frac{\text{Vs}}{\text{Am}}} =$$

$$H_L = \underline{\underline{915141 \frac{\text{A}}{\text{m}}}}$$

$$N \cdot I' = \sum H \cdot e$$

$$I' = \frac{H_E \cdot e_E + H_L \cdot e_L}{N} = \frac{550 \frac{A}{m} \cdot 0,46m + 915141 \frac{A}{m} \cdot 0,0005m}{800} =$$

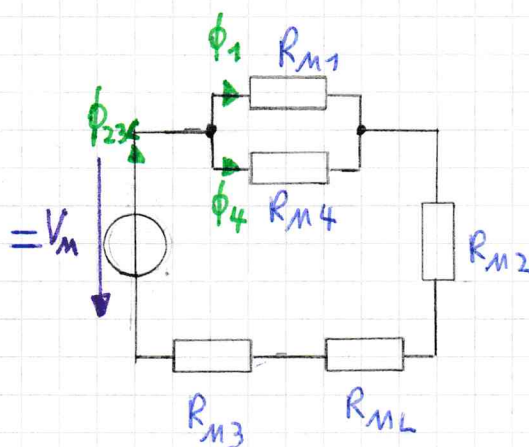
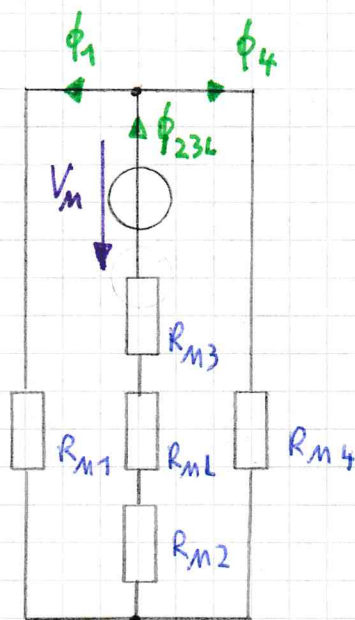
$$I' = \underline{\underline{0,888A}}$$



$$V_{ME} = H_E \cdot e_E = 550 \frac{A}{m} \cdot 0,46m = \underline{\underline{253A}}$$

$$V_{ML} = H_L \cdot e_L = 915141 \frac{A}{m} \cdot 0,0005m = \underline{\underline{457,6A}}$$

2)



a) $N = 20$ $I = 1A$

$\textcircled{H} = N \cdot I = \underline{\underline{20A}}$

b) $l_1 = l_4 = 10 \text{ cm} = \underline{\underline{10 \cdot 10^{-2} \text{ m}}}$ $\mu_0 = 4\pi \cdot 10^{-7} \frac{Vs}{Am}$
 $l_2 = l_3 = 3 \text{ cm} = \underline{\underline{3 \cdot 10^{-2} \text{ m}}}$ $\mu_{RE} = 1000$
 $l_L = 0,1 \text{ mm} = \underline{\underline{0,1 \cdot 10^{-3} \text{ m}}}$ $\mu_{RL} = 1$

$A_1 = A_4 = 1 \text{ cm}^2 = 1 \cdot (10^{-2} \text{ m})^2 = \underline{\underline{1 \cdot 10^{-4} \text{ m}^2}}$

$A_2 = A_3 = 4 \text{ cm}^2 = 4 \cdot (10^{-2} \text{ m})^2 = \underline{\underline{4 \cdot 10^{-4} \text{ m}^2}}$

$A_L = 1,1 \cdot A_2 = 4 \cdot 1,1 \cdot (10^{-2} \text{ m})^2 = \underline{\underline{4,4 \cdot 10^{-4} \text{ m}^2}}$

$R_m = \frac{l}{\mu_0 \cdot \mu_R \cdot A} \Rightarrow R_{m1} = \underline{\underline{795755 \frac{A}{Vs}}}$ mit μ_{RE}

$[R_m] = \frac{m}{\frac{Vs}{Am} \cdot m^2} = \frac{A}{Vs}$

$R_{m4} = R_{m1} = \underline{\underline{795755 \frac{A}{Vs}}}$ mit μ_{RE}

$R_{m2} = R_{m3} = \underline{\underline{59683 \frac{A}{Vs}}}$ mit μ_{RE}

$R_L = \underline{\underline{180858 \frac{A}{Vs}}}$ mit $\mu_{RL} \leftarrow \text{Achtung!}$

$$c) R_{M23L} = R_{M2} + R_{M3} + R_{ML} = \underline{\underline{300224 \frac{A}{Vs}}} \text{ (Serienschaltung)}$$

$$R_{M14} = \frac{R_{M1} \cdot R_{M4}}{R_{M1} + R_{M4}} = \frac{R_{M1}}{2} = \underline{\underline{397877 \frac{A}{Vs}}} \text{ (Parallelschaltung)}$$

↑ da $R_{M1} = R_{M4}$

$$R_{ges} = R_{M23L} + R_{M14} = \underline{\underline{698111 \frac{A}{Vs}}} \text{ (Serienschaltung)}$$

$$\textcircled{H} = V_M = R_M \cdot \phi$$

$$\phi_{ges} = \phi_{23L} = \frac{V_M}{R_{ges}} = \frac{20 A}{698111 \frac{A}{Vs}} = 29 \cdot 10^{-6} Vs = \underline{\underline{29 \cdot 10^{-6} Wb}}$$

$$\phi_1 = \phi \cdot \frac{R_{M4}}{R_{M1} + R_{M4}} = \frac{\phi}{2} = \underline{\underline{14,3 \cdot 10^{-6} Wb}}$$

(Stromteiler)

↑ da $R_{M1} = R_{M4}$

$$\phi_{ges} = \phi_1 + \phi_4 \Rightarrow \phi_4 = \phi - \phi_1 = \underline{\underline{14,3 \cdot 10^{-6} Wb}}$$

(Kirchhoff Knoten)

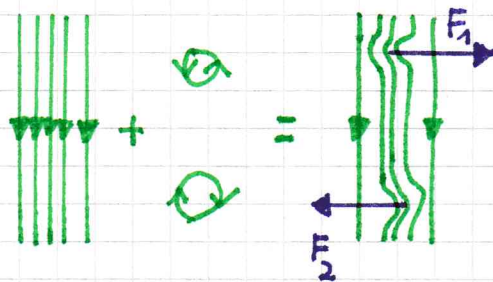
$$3) a) B = 100 \text{ mT} \quad I = 4 \text{ A}$$

$$\ell_B = 150 \text{ cm} \quad r_s = 100 \text{ cm}$$

$$\ell_R = 80 \text{ cm} \quad \ell_R = 10 \text{ cm}$$

$$\vec{F} = I (\vec{\ell} \times \vec{B}) = I \cdot B \cdot \ell = 4 \text{ A} \cdot 100 \text{ mT} \cdot 150 \text{ cm} =$$

$$= 4 \cdot 0,1 \cdot 1,5 \text{ ATm} = \underline{\underline{0,6 \text{ N}}} \quad \text{mit } T = \frac{\text{N}}{\text{Am}}$$



$$b) \quad M = F \cdot \ell = 1 \text{ Nm}$$

$$F = \frac{M}{\ell} = \frac{M}{\frac{\ell_R}{2}} = \frac{1 \text{ Nm}}{0,40 \text{ m}} = \underline{\underline{2,5 \text{ N}}}$$

$$F_1 = \frac{F}{2} = \underline{\underline{1,25 \text{ N}}}$$

$$F = B \cdot I \cdot \ell \Rightarrow I_1 = \frac{F_1}{B \cdot \ell} = \frac{1,25 \text{ N}}{0,1 \text{ T} \cdot 1,5 \text{ m}} =$$

$$I_1 = \underline{\underline{8,33 \text{ A}}}$$

$$I_2 = I_1$$

$$4 \quad F_L = F_Z$$

Lorentzkraft: $F_L = e \cdot v \cdot B$

Zentrifugalkraft: $F_Z = m_e \cdot \frac{v^2}{r}$

$$v = 10000 \frac{\text{km}}{\text{s}} = 10 \cdot 10^6 \frac{\text{m}}{\text{s}}$$

$$B = 50 \text{ mT} = 50 \cdot 10^{-6} \text{ T}$$

$$e = 1,602 \cdot 10^{-19} \text{ C}$$

$$m_e = 9,11 \cdot 10^{-31} \text{ kg}$$

$$e v B = m_e \frac{v^2}{r}$$

$$r = \frac{m_e \cdot v}{e \cdot B} = \underline{\underline{1,137 \text{ m}}}$$

$$[r] = \frac{\text{kg} \cdot \frac{\text{m}}{\text{s}}}{\text{C} \cdot \text{T}} = \frac{\text{kg} \cdot \frac{\text{m}}{\text{s}}}{\text{As} \cdot \frac{\text{kg}}{\text{As}^2}} = \underline{\underline{\text{m}}}$$